

Maven Interview Questions and Answers (3+ Years Experience)

1. What is Maven? Why is it used?

Maven is a build automation tool used for Java projects. It helps in compiling code, running tests, packaging into JAR/WAR, managing dependencies, and deploying to repositories using a pom.xml file.

2. What is a pom.xml?

pom.xml (Project Object Model) is the main config file of Maven. It defines project info, dependencies, plugins, and version details.

3. What are Maven build lifecycle phases?

Maven has predefined phases: validate, compile, test, package, install, deploy. Each step performs a specific task in the build process.

4. Difference between mvn clean install and mvn install?

mvn install builds and stores the package in local repo. mvn clean install first removes old builds, then does a fresh build and install.

5. What are the types of Maven repositories?

Local (.m2), Remote (e.g. Nexus), and Central (Maven Central). Local is your machine, others are external sources for dependencies.

6. Difference between Maven Central and Nexus?

Maven Central is a public repo with open libraries. Nexus is a private repo used by companies to manage internal builds and packages.

7. How do you add a dependency in Maven?

In pom.xml using <dependency> block. Maven downloads it automatically. Example: junit dependency for testing.

8. Difference between JAR, WAR, and EAR?

Maven Interview Questions and Answers (3+ Years Experience)

JAR: Java app, WAR: Web app, EAR: Enterprise app (JAR+WAR). WAR goes to web server, EAR to enterprise server.

9. What is the default location of the local Maven repo?

~/.m2/repository

10. What does mvn clean do?

Deletes previous build files in the target/ folder to ensure a fresh build.

11. What are Maven plugins?

Plugins help in compiling, testing, packaging. You can configure them in pom.xml. Examples: compiler plugin, surefire plugin.

12. What is Convention over Configuration in Maven?

Maven expects files in standard locations. If followed, no need for custom config. Example: src/main/java for code.

13. Difference between install and deploy?

install saves build locally in .m2. deploy uploads it to a remote repo (like Nexus).

14. Difference between compile and package?

compile: turns .java to .class, package: creates JAR/WAR from compiled files.

15. How to skip tests in Maven build?

Use: mvn install -DskipTests

16. What are the advantages of Maven?

Automates build, test, deploy. Manages dependencies. Standard structure. Easy to integrate with Jenkins and Git.